

Data Warehousing – Lab/Lec

Lesson 6 – Midterm Mini-Project

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Learning Objectives:

- **Knowledge (Cognitive):** Understand multi-engine architectural principles, relational star schemas, non-relational document stores, ELT data flows, and advanced OLAP querying constructs.
- **Skills (Psychomotor):** Design and code automated Python ETL scripts using pandas, sqlite3, and pymongo to cleanse, transform, load, and query heterogeneous datasets.
- **Attitudes (Affective):** Exhibit data stewardship, critical problem-solving in handling messy enterprise data anomalies, and collaborative engineering practices in team environments.

Group Mini-Project Titles & Domain Specifications:

Group 1: Global E-Commerce Sales & Customer Behavior Data Warehouse

- **Dataset:** [Online Sales Dataset on Kaggle](#)
- **Business Scenario:** An online retail marketplace requires an integrated data warehouse to track multi-regional transaction volumes, evaluate payment method impacts, and analyze customer purchasing trends.
- **Lab Description:** Build an ETL script to clean transaction logs, load structured records into an SQLite star schema (fact_sales), and store customer feedback logs in MongoDB. Execute window functions to track monthly revenue growth and MongoDB pipelines for category aggregations.
- **Expected Outcome:** A dual-engine database repository capable of powering executive sales dashboards and customer behavior reports.

Group 2: Supply Chain & Logistics Shipment Tracking Warehouse

- **Dataset:** [DataCo Supply Chain Dataset on Kaggle](#)
- **Business Scenario:** A global logistics enterprise needs to monitor delivery statuses, shipping delays, and inventory bottlenecks across international supply chain routes.
- **Lab Description:** Extract messy shipping logs, handle missing delivery dates and anomalous shipping costs via Python, load normalized shipments into SQLite, and store shipment sensor logs in MongoDB. Compute delivery delay trends using SQL window functions.
- **Expected Outcome:** An automated logistics data pipeline providing operational visibility into supply chain latency and fulfillment efficiency.

Group 3: Healthcare Patient Encounter & Treatment Cost Warehouse

- **Dataset:** [Healthcare Hospital Dataset on Kaggle](#) (Utilize standard hospital patient encounter open datasets)
- **Business Scenario:** A hospital network requires centralized analysis of patient admissions, treatment durations, and departmental billing costs to optimize resource allocation.
- **Lab Description:** Ingest multi-format patient records, cleanse null admission IDs and corrupted cost entries using Pandas, load structured billing into SQLite, and store unstructured clinical notes in MongoDB. Run analytical queries comparing patient length of stay across departments.
- **Expected Outcome:** A secure multi-engine storage model facilitating clinical cost analysis and resource utilization tracking.

Group 4: Fintech Digital Wallet Transactions & Fraud Log Warehouse

- **Dataset:** [Online Payments Fraud Detection Dataset on Kaggle](#) (Utilize standard financial transaction logs)
- **Business Scenario:** A digital financial services provider needs real-time tracking of transaction volumes, anomaly detection flags, and user wallet balances.
- **Lab Description:** Cleanse transaction amount anomalies and missing timestamps, load structured financial records into SQLite, and push flagged security alerts into MongoDB. Execute advanced window functions to calculate running user balances and fraud frequencies.
- **Expected Outcome:** A robust transaction data warehouse supporting financial auditing and risk mitigation reporting.

Group 5: Smart IoT Energy Consumption & Grid Metering Warehouse

- **Dataset:** [Household Electric Power Consumption Dataset on UCI/Kaggle](#)
- **Business Scenario:** A utility provider must process high-frequency smart meter readings and household power consumption logs to identify peak load times and grid anomalies.
- **Lab Description:** Handle missing sensor readings and malformed string metrics, load aggregated hourly usage into SQLite, and store raw telemetry payloads in MongoDB. Use analytical pipelines to compute rolling energy averages and peak consumption spikes.
- **Expected Outcome:** An analytical grid database enabling energy consumption trend forecasting and peak-load management.

Group 6: Hospitality Booking, Revenue, and Guest Feedback Warehouse

- **Dataset:** [Hotel Booking Demand Dataset on Kaggle](#)
- **Business Scenario:** A hotel chain requires integrated analytics on room reservations, cancellation rates, seasonal revenue trends, and unstructured guest review feedback.
- **Lab Description:** Cleanse missing guest demographic data and date strings, load reservation facts into SQLite, and store review text documents in MongoDB. Execute multi-faceted aggregation pipelines to correlate cancellation rates with room pricing.

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- **Expected Outcome:** A hospitality data warehouse optimized for occupancy rate forecasting and guest satisfaction analytics.

Group 7: EdTech Student Learning Engagement & Course Completion Warehouse

- **Dataset:** [Student Performance and Learning Analytics Dataset on Kaggle](#)
- **Business Scenario:** An online education platform needs to track student module completions, video watch times, quiz scores, and discussion forum interactions.
- **Lab Description:** Ingest disparate learning logs, clean null student IDs, load relational grade records into SQLite, and store clickstream activity logs in MongoDB. Apply window functions to track individual student progress over time.
- **Expected Outcome:** An academic data repository driving student retention analysis and learning path optimization.

Group 8: Retail Grocery Inventory & Point-of-Sale (POS) Warehouse

- **Dataset:** [Grocery Store Sales and Inventory Dataset on Kaggle](#)
- **Business Scenario:** A supermarket chain needs to monitor daily store sales, stock turnover rates, and inventory reorder thresholds across multiple branches.
- **Lab Description:** Process messy POS transaction logs, clean pricing currency symbols, load store sales into SQLite tables, and store supplier audit logs in MongoDB. Run monthly sales variance queries using SQL CTEs.
- **Expected Outcome:** A grocery warehouse model supporting automated inventory replenishment and stockout prevention.

Group 9: Telecommunications Network Traffic & Subscriber Usage Warehouse

- **Dataset:** [Telecom Customer Churn and Usage Dataset on Kaggle](#)
- **Business Scenario:** A telecommunications operator wants to analyze subscriber voice/data usage patterns, service plan changes, and customer churn risk factors.
- **Lab Description:** Ingest call detail records (CDRs), handle corrupted usage metrics, load structured subscriber metrics into SQLite, and store customer service interaction logs in MongoDB. Execute aggregation pipelines to segment high-usage accounts.
- **Expected Outcome:** A telecom data warehouse empowering customer retention strategies and network capacity planning.

Group 10: Real Estate Property Listing & Regional Market Value Warehouse

- **Dataset:** [House Sales Prediction Dataset on Kaggle](#)
- **Business Scenario:** A real estate brokerage requires a consolidated view of property listings, historical sale prices, square footage metrics, and regional market valuations.
- **Lab Description:** Cleanse missing property attributes and date formats, load property fact tables into SQLite, and store detailed architectural feature documents in MongoDB. Perform comparative price-per-square-foot trend queries.

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- **Expected Outcome:** A real estate analytical database facilitating automated property valuation and market trend reporting.

Group 11: Streaming Media User Subscription & Watch-Time Analytics Warehouse

- **Dataset:** [Netflix / Streaming TV Shows and Movies Dataset on Kaggle](#)
- **Business Scenario:** A digital streaming platform needs to evaluate user watch times, content popularity across genres, and subscription renewal patterns.
- **Lab Description:** Ingest streaming log dumps, clean malformed duration strings, load subscription records into SQLite, and store user watch history document logs in MongoDB. Use window functions to calculate user viewing streaks and top genre revenues.
- **Expected Outcome:** An entertainment analytics warehouse supporting content recommendation systems and subscriber engagement reporting.

MS Teams Submission Deliverables:

- **Milestone 1 (Due Sept 22, 2026):**
 - Upload etl_pipeline.py, the generated SQLite database (warehouse.db), and the Milestone 1 execution audit log (milestone1_log.txt) to the designated MS Teams assignment channel.
 - Screenshots of the expected output.
- **Milestone 2 (Due Sept 29, 2026):**
 - Upload the complete project repository including olap_analysis.py, final execution console logs (olap_output.txt), and the Group Analytical Summary Report (Group_X_Report.pdf) to MS Teams>Assignment
 - Screenshots of the expected output.

Rubrics:

Criteria	Excellent (4)	Proficient (2-3)	Developing (1)
ETL & Data Cleansing	Expertly handles duplicates, nulls, type-casting, and date standardizations across raw sources.	Cleans primary anomalies but misses minor edge cases.	Fails to transform data or crashes due to unhandled exceptions.
Multi-Engine Loading	Successfully populates both relational SQLite tables and MongoDB document collections without constraint violations.	Loads data into both engines with minor schema or mapping discrepancies.	Fails to load data into one or both storage targets.
Advanced OLAP Querying	Implements robust SQLite CTEs, window functions (LAG, SUM OVER), and MongoDB \$facet pipelines accurately.	Writes functional analytical queries with minor syntax errors or suboptimal framing.	Relies on invalid query syntax or omits required analytical constructs.
Code Quality & Documentation	Code is modular, thoroughly commented, version-controlled, and accompanied by detailed execution logs and documentation.	Code executes successfully with basic comments and standard logs.	Code is unreadable, poorly structured, or missing execution proof.

Date: September 15, 2026